

Section 1, Question 1

Matrix Multiplication with MapReduce

Distributed Systems — Homework 3

1 What the question asks

Multiply two matrices, A of size $m \times n$ and B of size $n \times p$, using the MapReduce programming model: independent map tasks running on separate machines, a shuffle that groups related values together, and a reduce step that produces the final matrix.

The implementation must handle arbitrary matrix shapes, including cases where the number of rows does not divide evenly by the number of tasks, and must produce the same answer regardless of how many tasks are used.

2 How the multiplication is decomposed

To compute one output cell $C[i][j]$ you need row i of A and column j of B . The decomposition rests on a simple observation: if a map task is given **whole rows of A** and can read **all of B** , then it can compute every cell in its own rows without consulting any other task. That independence is what makes the work splittable.

Map. For each row i of A held by this task, and each column j of B , emit i,j <TAB> partial value.

Combine. A task usually produces several partials for the same cell. The combiner adds them locally, so less data crosses the network.

Shuffle. `sort -k1,1` places all values for one cell together. No program performs the shuffle — the sort is the shuffle.

Reduce. Walk the sorted stream, sum the values sharing a key, print the finished matrix.

The pipeline is therefore five stages: split A by rows, map, sort per machine, combine, sort globally, reduce. Splitting by rows is what permits any task count: a task that receives no rows simply produces nothing, which is why a matrix with 1001 rows divided among four tasks works exactly as well as one with 1000.

3 What was implemented

File	Role	Lines
<code>mapper.py</code>	emits one partial product per output cell	45
<code>combiner.py</code>	sums a task's own partials before they travel	30
<code>reducer.py</code>	sums partials per cell and prints the matrix	40
<code>verify.py</code>	multiplies directly, with no MapReduce — the oracle	35
<code>run_cluster_test.sh</code>	Slurm: the nine correctness cases	
<code>benchmark_dist_slurm.sh</code>	Slurm: seven matrix shapes at four machines	
<code>benchmark_scaling_slurm.sh</code>	Slurm: three sizes $\times$ 1, 2, 4, 7 machines	

Everything is Python 3, which the compute nodes already provide, so there is no build step.

4 Correctness

Nine cases, all passing, run distributed across four machines. Each result is compared against `verify.py`, which performs the multiplication directly.

The cases are chosen so that each could plausibly fail for a different reason: the assignment’s worked example; a square matrix whose rows divide evenly by the task count and one that does not (1001 rows over four tasks); a tall matrix (5000×10); a wide one (10×500), where most tasks receive almost nothing; and the degenerate single-row and single-column cases.

The scaling benchmark adds a stronger check: **all 36 of its runs** were compared against the single-task answer, confirming that dividing the input differently never changes the result.

5 Results

5.1 Scaling: the larger the input, the better it scales

Median of three runs, one map task per machine.

Input	1 machine	2	4	7	Speedup at 7
Small (100×50)	0.414 s	0.353 s	0.355 s	0.326 s	$1.27\times$
Medium (500×50)	1.141 s	0.720 s	0.529 s	0.446 s	$2.56\times$
Large (2000×50)	3.801 s	2.140 s	1.274 s	0.918 s	$4.14\times$

The pattern is the finding. Large reaches $4.14\times$ on seven machines (59% efficiency), Medium $2.56\times$ (37%), and Small barely moves at $1.27\times$ (18%).

Every run carries a fixed cost that does not shrink when machines are added: starting a Python interpreter on each node, reading all of B on every task, and five stages of process startup. On a 100×50 matrix there is so little real work that this fixed cost is most of the runtime, so the extra machines have almost nothing to divide.

The practical conclusion is that **small inputs should not be distributed**. One machine finishes sooner than seven once coordination is counted. This is the same conclusion Section 2 reaches for a completely different workload, which suggests it is a property of the model rather than of either program.

5.2 Shape matters more than size

Seven matrix shapes, all on four machines.

Shape	A	B	Map	Shuffle	Total
wide	10×500	500×10	0.22	0.19	0.65 s
small	100×50	50×50	0.26	0.20	0.71 s
tall	5000×10	10×10	0.34	0.24	0.93 s
medium	500×50	50×50	0.45	0.23	1.07 s
large	2000×50	50×50	1.15	0.42	2.45 s
square	1000×100	100×1000	17.53	20.55	50.22 s
square uneven	1001×100	100×1000	18.34	22.99	53.96 s

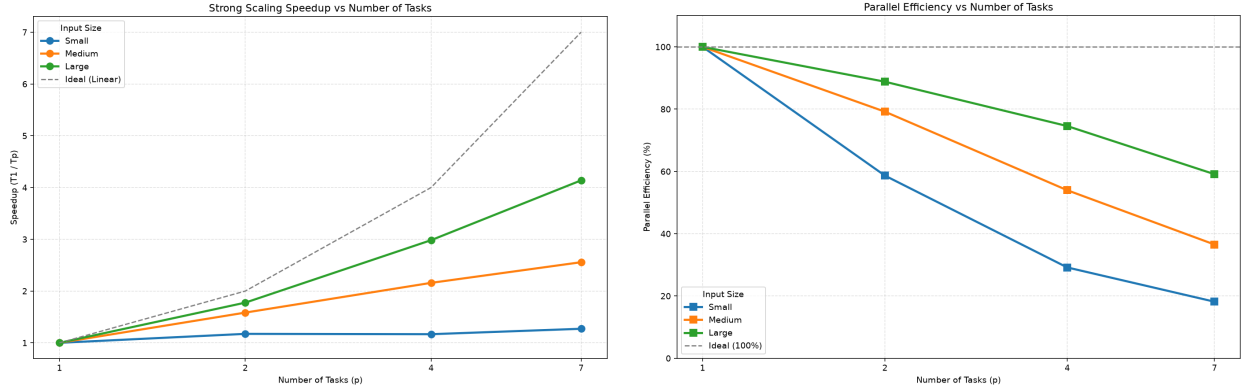


Figure 1: Strong scaling. Left: speedup against the linear ideal — the three input sizes separate clearly, and only Large stays near the ideal. Right: the same data as parallel efficiency, which falls for every size but far faster for the small inputs.

Square is twenty times slower than large, despite having half as many rows. The reason is the number of output cells: `large` produces $2000 \times 50 = 100,000$ of them, while `square` produces $1000 \times 1000 = 1,000,000$, and each requires 100 partial products. The cost follows $m \times n \times p$, not the size of A — so the row count, which is what determines how the work is split, is a poor predictor of how long a run will take.

The shuffle becomes the bottleneck at scale. For `square`, sorting takes 20.5s of the 50.2s total — more than the mapping itself. For every other shape it is under half a second. This is the characteristic weakness of MapReduce: intermediate data must be materialised and sorted, and that cost grows with the number of distinct keys, which here is the number of output cells.

Uneven splits cost very little. 1001 rows over four tasks takes 53.96s against 50.22s for 1000 rows. The single extra row lands on one machine and the other three wait at the barrier — a 7% penalty, which is the expected behaviour of a synchronisation point rather than a defect.

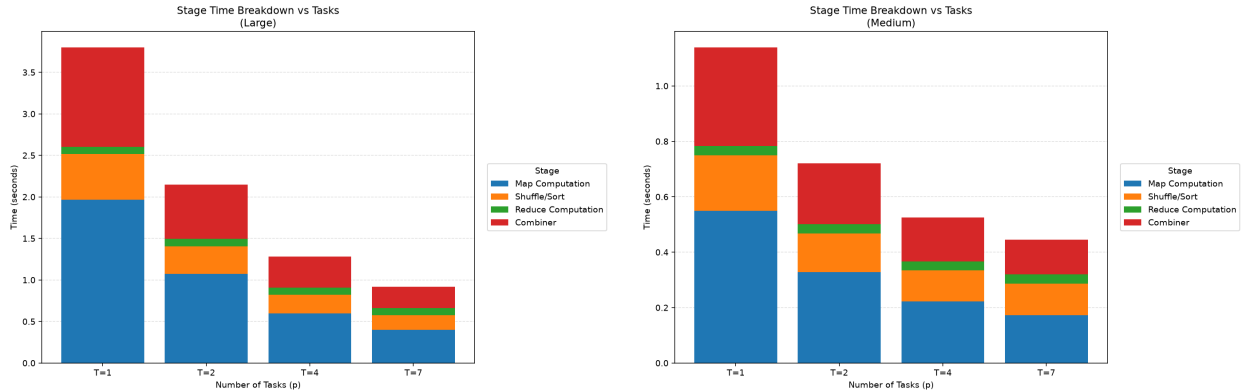


Figure 2: Where the time is spent, for the Large and Medium inputs. The map stage shrinks as machines are added, while the shuffle and the single reduce do not, which is what limits the achievable speedup.

6 A measurement trap worth recording

Only seven of the cluster's eight nodes are usable. An earlier sweep nonetheless requested eight map tasks. The allocation grants one task slot per machine, so `srun --ntasks=8` across seven machines quietly started only *seven* tasks: one chunk of A was never mapped.

The pipeline then produced a **wrong answer faster than the correct one** — 0.308s at the nominal eight tasks against 0.327s at four. A missing chunk is indistinguishable from a speedup

if only the clock is read.

Two changes followed. The sweep now runs 1, 2, 4 and 7 tasks, every point strictly one task per machine; and the benchmark verifies that the number of map outputs equals the number of chunks, stopping if they differ. The lesson generalises: a performance result that is not accompanied by a correctness check is not a result.

7 Conclusions

Matrix multiplication decomposes cleanly into MapReduce because each map task can compute whole rows of the answer independently, given all of B . The implementation is correct across all nine test cases and across every task count tried.

Speedup depends strongly on having enough work to divide: $4.14\times$ on seven machines for the largest input, but only $1.27\times$ for the smallest, where fixed coordination costs dominate. Shape matters more than size, because cost follows the number of output cells rather than the number of rows — and for the squarest case the shuffle, not the computation, becomes the limiting stage.